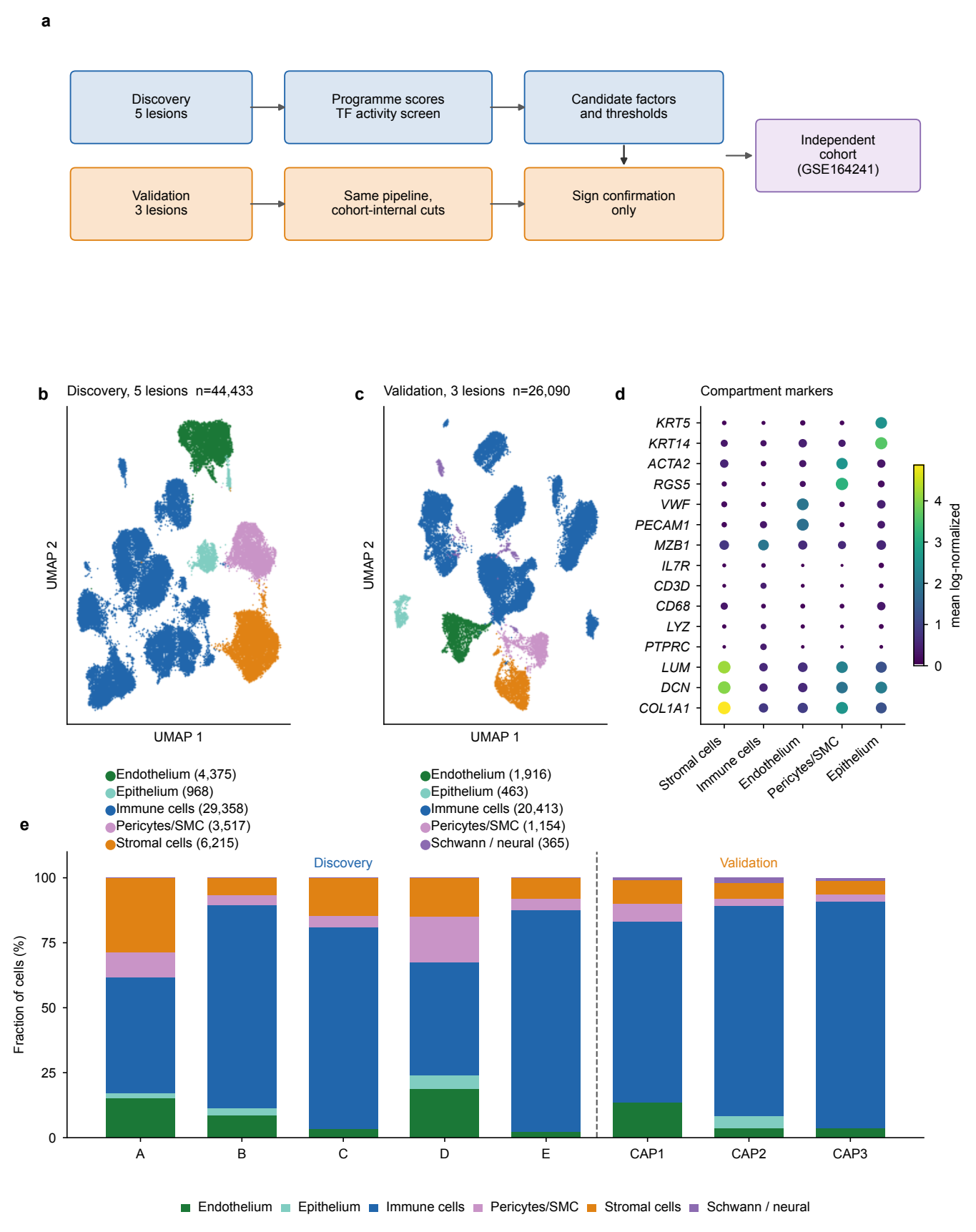

a
Discovery
5 lesions
Programme scores
TF activity screen
Candidate factors
and thresholds
Independent
cohort
(GSE164241)
Validation
3 lesions
Same pipeline,
cohort-internal cuts
Sign confirmation
only
b
Discovery, 5 lesions n=44,433
UMAP 2
UMAP 1
Endothelium (4,375)
Epithelium (968)
Immune cells (29,358)
Pericytes/SMC (3,517)
Stromal cells (6,215)
c
Validation, 3 lesions n=26,090
UMAP 2
UMAP 1
Endothelium (1,916)
Epithelium (463)
Immune cells (20,413)
Pericytes/SMC (1,154)
Schwann / neural (365)
d
Compartment markers
KRT5
KRT14
ACTA2
RGS5
VWF
PECAM1
MZB1
IL7R
CD3D
CD68
LYZ
PTPRC
LUM
DCN
COL1A1
Stromal cells
Immune cells
Endothelium
Pericytes/SMC
Epithelium
mean log-normalized
0
1
2
3
4
e
Discovery
Validation
Fraction of cells (%)
0
25
50
75
100
A
B
C
D
E
CAP1
CAP2
CAP3
Endothelium
Epithelium
Immune cells
Pericytes/SMC
Stromal cells
Schwann / neural